

# PYROLYSIS HEATING JACKET

## DESIGN OPTIMIZATION SUMMARY

Generated: 12-Apr-2026 | Project: PY-301

### FIXED OPERATING CONDITIONS:

|                           |                                |
|---------------------------|--------------------------------|
| Salt mass flow rate:      | 0.7609 kg/s (Ternary Eutectic) |
| Pyrolizer diameter:       | 318.0 mm                       |
| Heating zone length (L1): | 2.05 m                         |
| Constant temp zone (L2):  | 2.57 m                         |
| Salt inlet temperature:   | 611.72 C                       |
| Wall inlet temperature:   | 280 C                          |
| Wall outlet temperature:  | 550 C                          |
| Heat duty (Section 1):    | 23550 W                        |
| Heat duty (Section 2):    | 20070 W                        |

### DESIGN COMPARISON

| Parameter           | Baseline | Non-Finned Optimal | Finned Optimal | Units  |
|---------------------|----------|--------------------|----------------|--------|
| Status              | INVALID  | VALID              | VALID          | -      |
| D_jacket            | 0.358    | 0.3397             | 0.500          | m      |
| GAP (Annular Gap)   | 20.0     | 10.84              | 91.0           | mm     |
| N_ch (Channels)     | 10       | 8                  | 7              | -      |
| t_fin (Wall Thick.) | 5.0      | 14.65              | 2.0            | mm     |
| Longitudinal Fins   | 0        | 0                  | 9              | per ch |
| Fin Height          | -        | -                  | 18.16          | mm     |
| Fin Thickness       | -        | -                  | 4.71           | mm     |
| Area Enhancement    | 1.0x     | 1.0x               | 3.01x          | -      |
| Pressure Drop       | 28.73    | 159.96             | 2.58           | kPa    |
| DP Change vs Base   | -        | +457%              | -91%           | -      |
| Velocity            | 19.56    | 38.84              | 4.41           | mm/s   |
| h2 (Heat Trans.)    | 123.9    | 184.9              | 61.5           | W/m²K  |
| S2 Area Margin      | -49.3    | 0.0003             | 0.07           | %      |

### DESIGN DESCRIPTIONS:

Baseline: Original design with GAP=20mm, N\_ch=10, no longitudinal fins.  
FAILED: Section 2 area margin = -49.3% (insufficient heat transfer area).  
Pressure drop = 28.73 kPa.

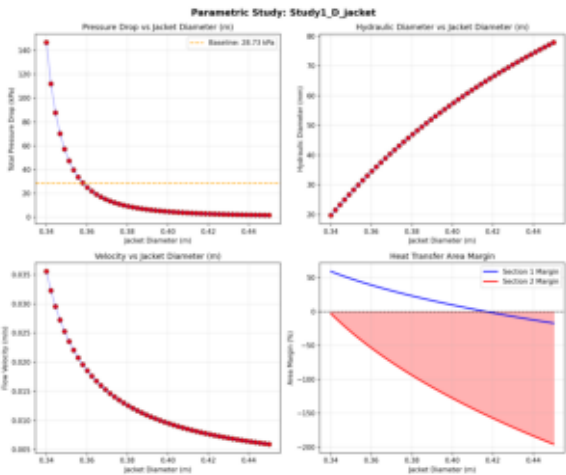
Non-Finned Optimal: Minimum DP design WITHOUT longitudinal fins.  
Required smaller gap (10.8mm) to achieve sufficient h2 for Section 2.  
VALID but DP increased to 159.96 kPa (+457% vs baseline).

Finned Optimal: Minimum DP design WITH longitudinal fins (rib geometry).  
9 fins/channel (18.2mm height) increase effective area by 3.01x.  
Allows larger gap (91mm), lower velocity, VALID with DP = 2.58 kPa (-91%).

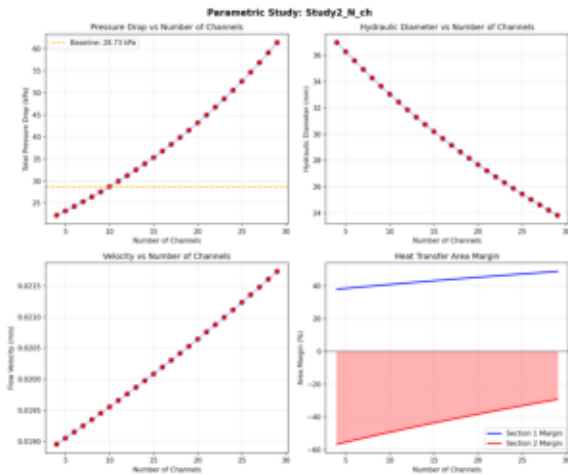
# STUDY 1: PARAMETRIC OPTIMIZATION WITHOUT FINS

Exploring GAP, N\_ch, and t\_fin to achieve valid design

## Effect of Jacket Diameter



## Effect of Number of Channels



## Baseline vs Non-Finned Optimal

## Valid Design Analysis

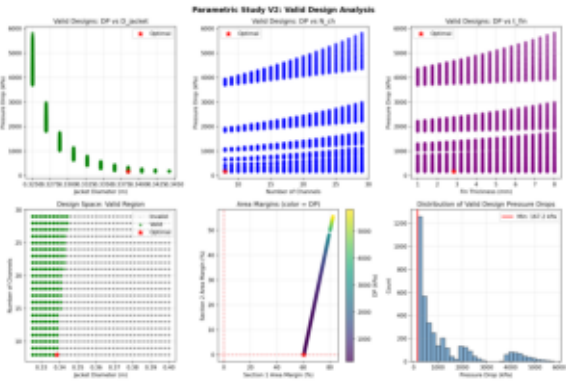


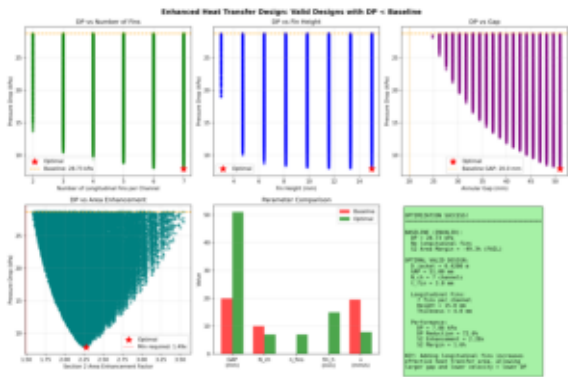
Image not found:  
baseline\_vs\_optimal.png

- KEY FINDINGS - WITHOUT FINS:
- Baseline fails because Section 2 requires  $A_{req} = 3.83 \text{ m}^2$  but only  $A_{av} = 2.57 \text{ m}^2$  available
  - To achieve valid design, h2 must increase by 49% (requires smaller gap, higher velocity)
  - Minimum achievable DP for valid non-finned design: 159.96 kPa (+457% vs baseline)

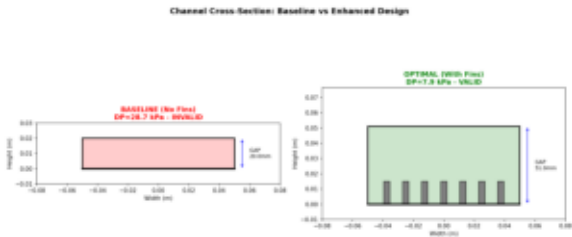
# STUDY 2: OPTIMIZATION WITH LONGITUDINAL FINS

Adding fins to increase effective heat transfer area

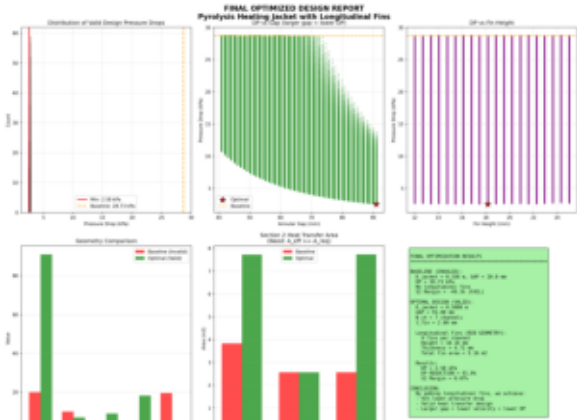
## Enhanced Design Analysis



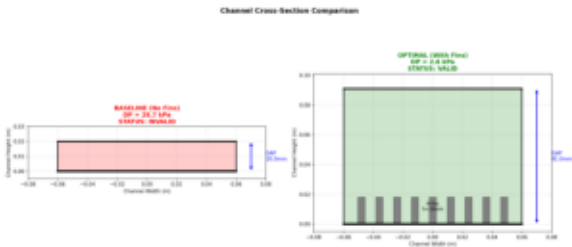
## Channel Cross-Section



## Final Optimization Report



## Final Cross-Section Comparison

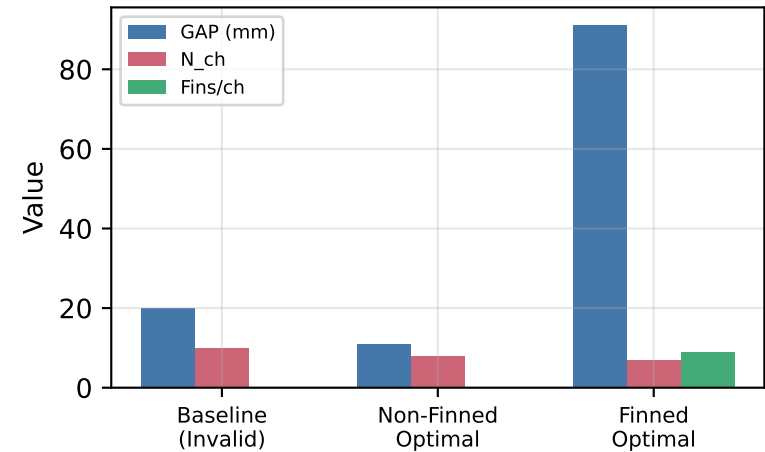


### KEY FINDINGS - WITH FINS:

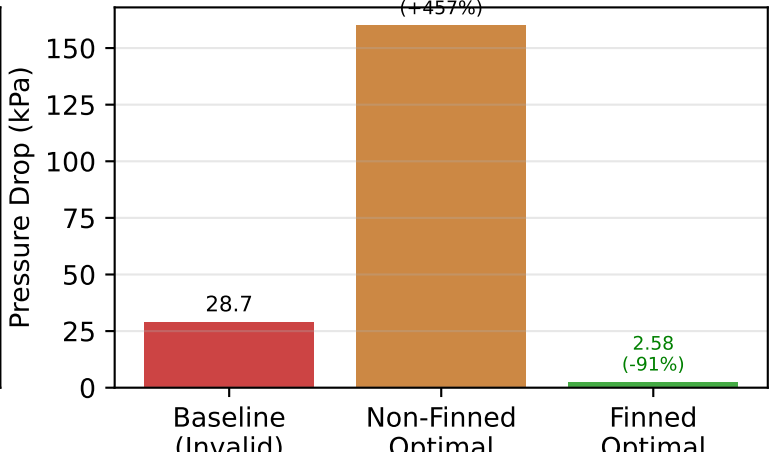
- Longitudinal fins increase effective heat transfer area by 3.01x
- Larger gap possible (91mm vs 20mm baseline) → lower velocity → lower DP
- Achieved 91% pressure drop reduction (28.73 kPa → 2.58 kPa) with valid design

# DESIGN RECOMMENDATIONS

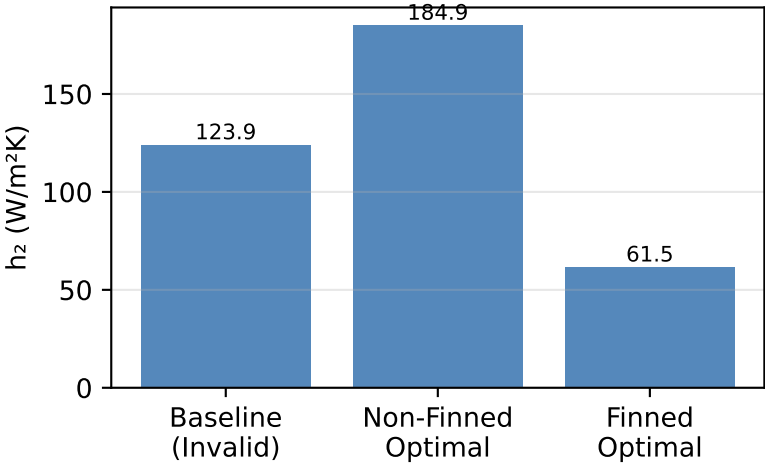
### Geometry Parameters



### Pressure Drop Comparison

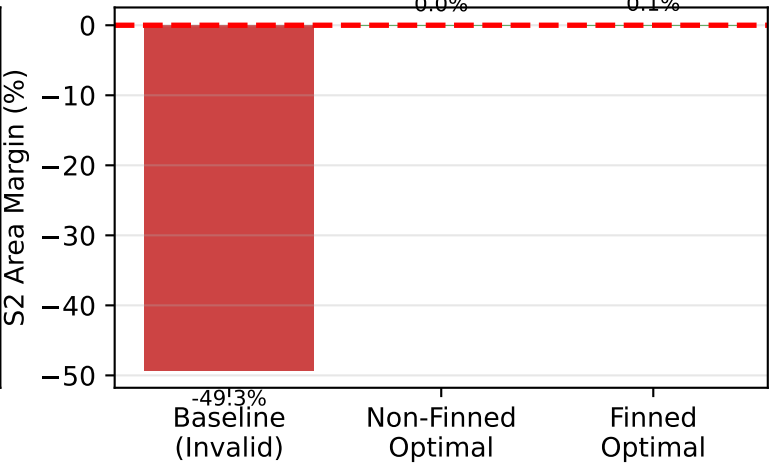


### Heat Transfer Coefficient (Section 2)

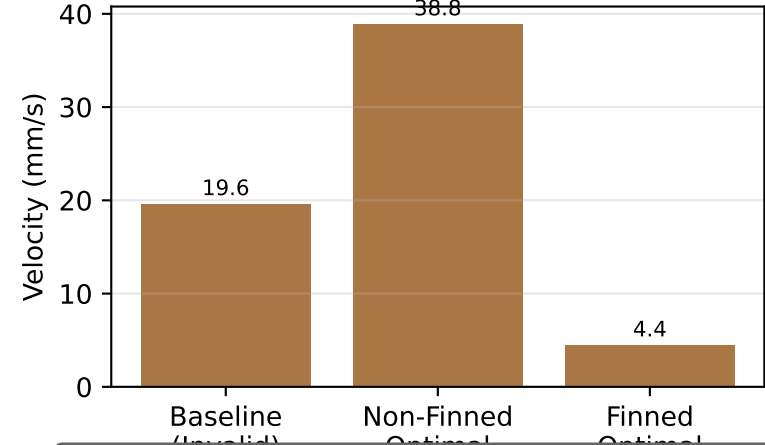


### Section 2 Area Margin

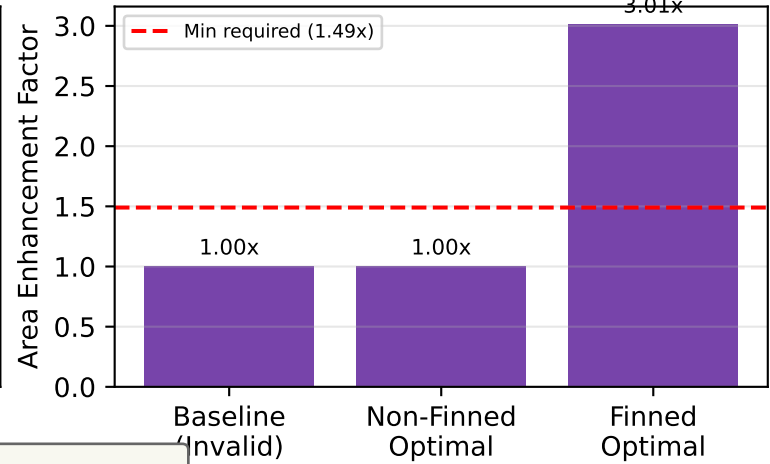
(Must be  $\geq 0$  for valid design)



### Flow Velocity



### Effective Area Enhancement



#### RECOMMENDATIONS:

##### ✓ CHOOSE FINNED OPTIMAL DESIGN (Recommended):

- Achieves 91% pressure drop reduction vs baseline (2.58 kPa vs 28.73 kPa)
- Valid heat transfer design with 0.07% safety margin on Section 2
- Requires 9 longitudinal fins per channel (18.2mm height, 4.7mm thick)
- Larger jacket diameter (500mm) accommodates increased gap

##### ✗ AVOID BASELINE DESIGN:

- INVALID: Section 2 area margin is -49.3% (insufficient heat transfer)

##### ✗ AVOID NON-FINNED OPTIMAL:

- Although valid, pressure drop is 5.6x higher than baseline (159.96 kPa)
- Only use if fins cannot be manufactured

**KEY INSIGHT:** Longitudinal fins provide the additional heat transfer surface needed without requiring high velocities, enabling both valid design AND minimum DP.